

Clinical Risk Assessment Report: Patient 89

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 18.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.82x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT), which reduced the patient's absolute risk by 2.2%. Biologically, this feature quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Large Zone Emphasis (CT): A decrease would alter the prediction outcome.
- Tumor Sphericity (CT): A decrease would alter the prediction outcome.
- Patient Age: A decrease would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.31 [Bottom 22%] (+2.5%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.05 [Avg (51th %ile)] (+2.1%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.69 [Avg (69th %ile)] (+1.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.2 [Bottom 23%] (+0.8%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.0 [Top 16%] (-2.2%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

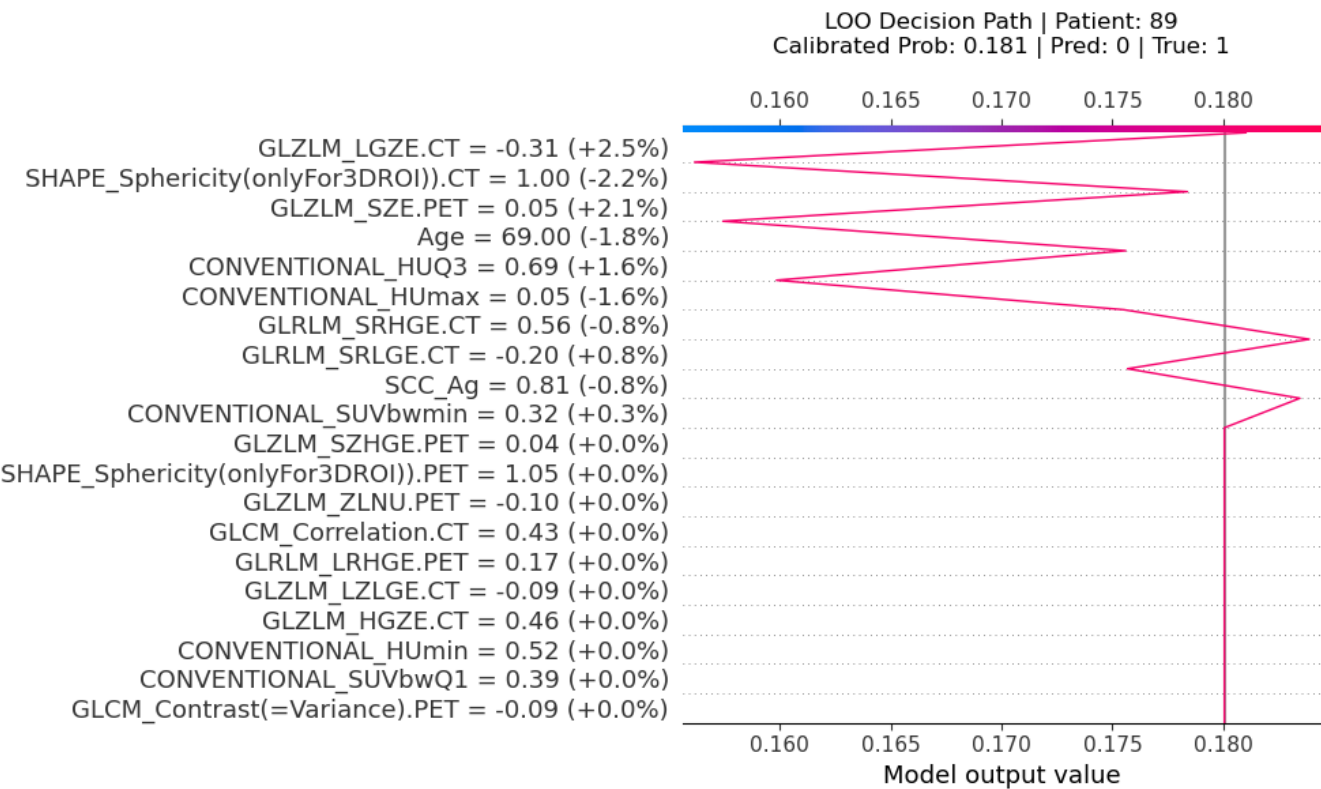
Patient Age (Age) **Value: 69.0 [Avg (73th %ile)] (-1.8%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.05 [Avg (47th %ile)] (-1.6%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) Value: 0.56 [Avg (63th %ile)] (-0.8%)

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

